

Dr Waseem Raza, PhD

Machine Learning Engineer | Applied AI & Data Science

Applied Machine Learning | Data Science | Model Evaluation | Intelligent Automation

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Professional Summary

Machine Learning Engineer and Data Scientist with a PhD in Electrical and Computer Engineering and 7+ years of experience developing data-driven systems across enterprise, industry, and research environments. Experience includes LLM-enabled analytics, AI-ready data pipelines, domain-adaptive machine learning, and scalable cloud solutions using Python, SQL, Azure Databricks, Apache Spark, and Azure services. At AT&T, lead development and operational integration of AI solutions for critical service-assurance workflows, improving automated diagnostic accuracy from 20% to 80%. Combine model evaluation, data quality, ETL, and cross-functional delivery to translate technical requirements into production-oriented AI capabilities.

Technical Expertise, Leadership, and Professional Strengths

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, transfer learning, domain adaptation, adversarial learning, anomaly detection, classification, regression, time-series forecasting, cross-validation, and model evaluation

Data Engineering: ETL/ELT pipelines, AI-ready datasets, batch processing, ingestion, cleansing, schema transformation, validation, feature engineering, data-quality assessment, and structured and unstructured data workflows

Data & AI Platforms: Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Blob Storage, Azure Data Lake, Google BigQuery, and cloud analytics workflows

Programming & Analytics: Python, SQL, C/C++, and R; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter, and exploratory data analysis

Generative & Agentic AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering, document ingestion, chunking, embeddings, semantic retrieval, structured information extraction, tool-using agents, agent orchestration, and grounded response generation

LLM Frameworks & Models: Hugging Face Transformers, LangChain, LangGraph, OpenAI APIs, transformer architectures, fine-tuning concepts, NLP foundation models, prompt templates, structured outputs, and LLM evaluation

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, shell scripting, model deployment, reproducible workflows, and cloud-native application development

Microsoft AI Ecosystem: Microsoft Foundry, Azure OpenAI, Azure AI Services, Azure AI Search, Azure Machine Learning, Foundry models and agents, Azure Content Understanding, Semantic Kernel, and responsible AI concepts

Leadership and Professional Strengths

- Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer

Impac Exploration Services

Sep. 2024-Dec. 2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.

- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Advanced-Degrees Intern
AT&T Labs

Jun. 2024-Aug. 2024
Austin, TX

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern
Futurewei Technologies

Sep. 2022-Jan. 2023
Schaumburg, IL

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern
AT&T Labs

Jun. 2022-Aug. 2022
Bedminster, NJ

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant
AI4Networks, University of Oklahoma

Sep. 2019-Sep. 2024
Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	09/2019-07/2025
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	08/2014-08/2016
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	11/2010-06/2014

Selected Projects

Financial ML and Forecasting <i>Portfolio Project</i>	2024-2025
• Credit-risk and loan-default modeling alongside time-series forecasting workflows for financial data. <i>Tools & methods:</i> Python, Machine Learning, Time Series, Model Evaluation	
Wearable Intelligence for Health Monitoring <i>Research Project</i>	2024-2026
• Deep-learning methods for continuous recognition of medication and dietary activities from wrist-worn inertial sensors. <i>Tools & methods:</i> Wearable AI, Time Series, Deep Learning, Smartwatch Sensing	
Cloud Computer Vision <i>Guided Project</i>	2025
• Real-time object detection, vehicle counting, and image-classification workflows using cloud-native AI services. <i>Tools & methods:</i> Computer Vision, Streaming, Google Cloud AutoML, Python	
AI-Ready Lakehouse Pipeline <i>Portfolio Project</i>	2024
• An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads. <i>Tools & methods:</i> Azure, Databricks, Spark, Data Engineering	

Certifications and Continuing Education

1. Machine Learning — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization

2. Google Cloud Data Analytics Professional Certificate — Google Cloud / Coursera | Completed

Focus: Cloud data transformation, analysis, visualization, and decision support using Google Cloud tools.

Tools & Skills: SQL, BigQuery, Data Analytics, Visualization

3. IBM Generative AI Engineering Professional Certificate — IBM / Coursera | Completed (13/16) courses

Focus: Sixteen-course professional certificate developing job-ready skills across AI foundations, Python application development, machine learning, deep learning, NLP, transformer architectures, LLM fine-tuning, RAG, LangChain, and AI agents.

Tools & Skills: Generative AI Engineering, Large Language Models, Transformer Architectures, Prompt Engineering, RAG, LangChain, AI Agents, Hugging Face, PyTorch, Keras, Scikit-learn, Python, Flask, Gradio, PEFT, LoRA, QLoRA, RLHF, DPO, PPO, NLP, Machine Learning, Deep Learning, Vector Databases

4. Deep Learning for Healthcare Specialization — [University of Illinois Urbana-Champaign / Coursera](#) | Completed (0/3) courses

Focus: An advanced specialization covering health-data processing, neural-network methods, and the development of deep-learning solutions for real-world medical and healthcare applications.

Tools & Skills: Deep Learning, Healthcare AI, Health Informatics, Health Data Processing, Artificial Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, Autoencoders, Generative Models, Medical Image Analysis, Supervised Learning, Unsupervised Learning, Predictive Modeling, Model Evaluation, PyTorch, Jupyter Notebooks

5. Machine Learning and Reinforcement Learning in Finance Specialization — [New York University / Coursera](#) | Completed (0/4) courses

Focus: An applied machine-learning specialization focused on financial prediction, portfolio management, trading, risk modeling, market dynamics, and reinforcement-learning methods for finance.

Tools & Skills: Machine Learning for Finance, Reinforcement Learning, Financial Modeling, Portfolio Management, Financial Trading, Risk Modeling, Predictive Modeling, Supervised Learning, Unsupervised Learning, Statistical Machine Learning, Market Dynamics, Derivatives, TensorFlow, Scikit-learn, Python

Selected Publications

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin

2024

IEEE PIMRC